Lab 7: Fuzz Testing

House Painting Quote - Input Validation

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Repository: github.com/swarapotd-rgb/L6_L7_63_house_painting_quote (LAB7 folder)

Objective and method

Evaluate whether the three parser functions handle malformed inputs without crashing. Each Hypothesis test uses `st.text() | st.none()` to generate strings and None, calls one function, and reports any exception as a test failure. The same tests are used before and after the fixes.

Function	Problem found	Fix applied
clean_colour_name	None causes AttributeError in strip().	Check string type and reject blank names.
room_area	Malformed dimensions / None cause errors.	Check type and format; catch conversion errors; require positive dimensions.
total_wall_area	Empty or invalid entries / None cause errors.	Check type; handle invalid numbers; reject negative areas.

Results and interpretation

Before the fixes: **3 tests failed**. After the fixes: **3 tests passed**. Invalid inputs return None; the interactive room-size prompt displays an error and asks again. The fixed run rejects abc, calculates 12x10 as 120 sq ft, and totals 120;95;110 as 325 sq ft.

Scope and reflection status

Passing these tests supports crash resistance for the generated strings and None; it does not prove correctness for every input or measure performance. Three official reflection questions were referenced in the README but were not supplied. This report records the observed bugs, fixes and learning; official question-specific answers remain pending.

SS1 - Original application failure

Command: py house_painting_quote_parser.py

```
PS C:\Users\swara\L6_L7_63_house_painting_quote\LAB7> py house_painting_quote_parser.py
==== Running clean_colour_name ====
Enter a paint colour name (e.g. Ocean Blue): Ocean Blue
Colour code: ocean_blue

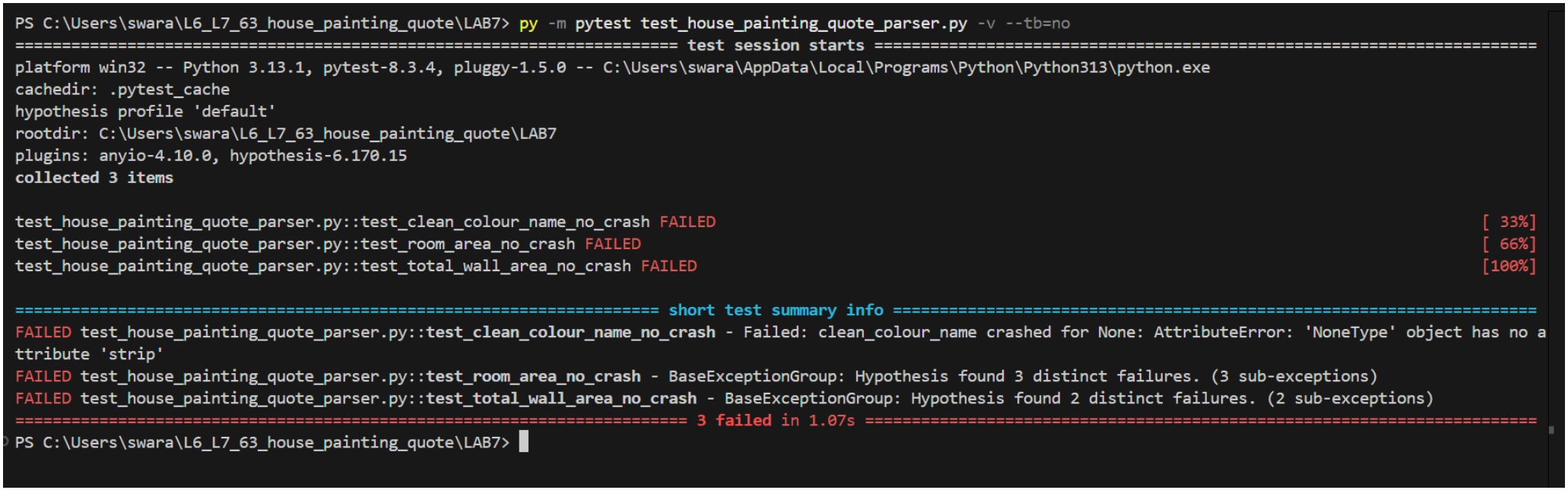
==== Running room_area ====
Enter room size as LENGTHxWIDTH in feet (e.g. 12x10): abc
Traceback (most recent call last):
  File "C:\Users\swara\L6_L7_63_house_painting_quote\LAB7\house_painting_quote_parser.py", line 37, in <module>
    main()
    ~~~~^^
  File "C:\Users\swara\L6_L7_63_house_painting_quote\LAB7\house_painting_quote_parser.py", line 30, in main
    print("Room area (sq ft):", room_area(value))
                                ~~~~~~~~~^^^^^^^
  File "C:\Users\swara\L6_L7_63_house_painting_quote\LAB7\house_painting_quote_parser.py", line 13, in room_area
    length, width = text.lower().split('x')
    ^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS C:\Users\swara\L6_L7_63_house_painting_quote\LAB7>
```

Observation

Ocean Blue is converted to ocean_blue. Entering abc as the room size triggers a ValueError because the split does not produce two dimensions. The original program terminates before asking for wall areas.

SS2 - Fuzz tests before fixes

Command: py -m pytest test_house_painting_quote_parser.py -v --tb=no

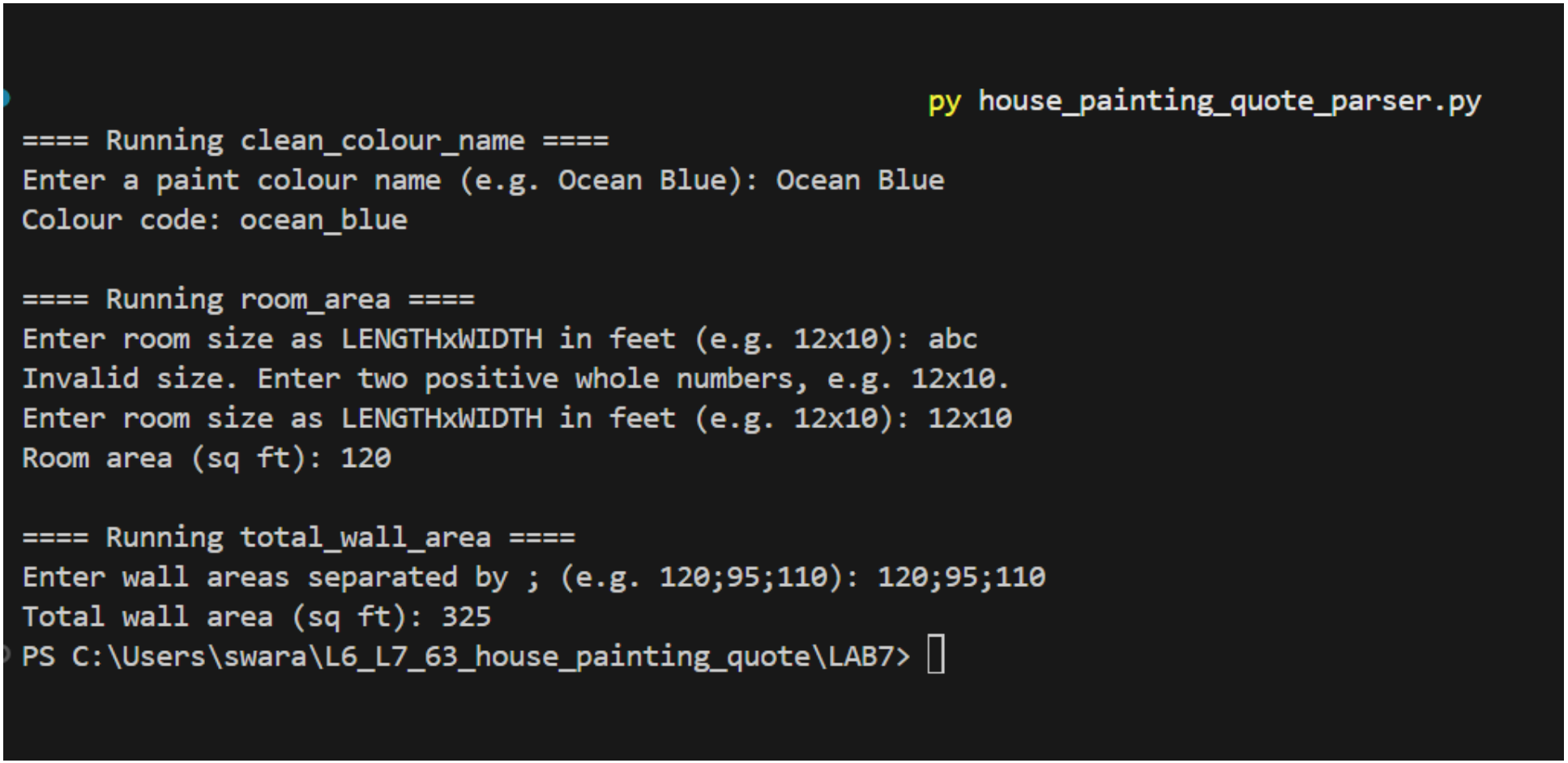


Observation

All three property tests fail (3 failed in 1.07s). The screenshot records None causing a colour-name error and Hypothesis reporting multiple failure types in the room and wall parsers. The compact output omits full tracebacks.

SS3 - Fixed application behaviour

Command: py house_painting_quote_parser.py



Observation

The application rejects abc with a clear message and requests another room size. The valid replacement 12x10 returns 120 sq ft. Execution continues, and wall inputs 120;95;110 total 325 sq ft.

SS4 - Fuzz tests after fixes

Command: py -m pytest test_house_painting_quote_parser.py -v --tb=short

```
PS C:\Users\swara\L6_L7_63_house_painting_quote\LAB7> py -m pytest test_house_painting_quote_parser.py -v --tb=short
============================================= test session starts =============================================
platform win32 -- Python 3.13.1, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\swara\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\swara\L6_L7_63_house_painting_quote\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 3 items

test_house_painting_quote_parser.py::test_clean_colour_name_no_crash PASSED                          [ 33%]
test_house_painting_quote_parser.py::test_room_area_no_crash PASSED                                  [ 66%]
test_house_painting_quote_parser.py::test_total_wall_area_no_crash PASSED                            [100%]

============================================== 3 passed in 0.70s ==============================================
PS C:\Users\swara\L6_L7_63_house_painting_quote\LAB7>
```

Observation

All three property tests pass (3 passed in 0.70s). Each test checks one function across generated inputs; three passing tests does not mean only three individual input values were tested.

Environment shown: Windows, Python 3.13.1, pytest 8.3.4, Hypothesis 6.170.15. Timings describe these individual runs and are not a performance benchmark.